

Strain-Controlled Atomic Reconstruction and Quasi-1D Excitons in Moiré Heterostructures

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In two-dimensional materials, strain provides effective means for tailoring electronic and optical properties. While uni- or biaxial strain has been widely implemented in monolayer semiconductors, deterministic control over atomic reconstruction and the resulting microscopic stacking textures in moiré heterostructures remains challenging. Here, we demonstrate the controlled formation of one-dimensional quantum wire arrays in MoSe₂-WSe₂ heterobilayers through the interplay of uniaxial strain and atomic reconstruction. This process yields one-dimensional confinement of interlayer excitons within domain walls, producing near-unity linearly polarized emission due to confinement-induced symmetry breaking. The domain wall width and thereby the degree of exciton confinement can be precisely tuned via the interlayer twist angle. Under an out-of-plane electric field, the confined excitons exhibit Stark shifts exceeding 100 meV and fine-structure splitting modulations by up to a factor of 2. These findings establish strain tuning as a powerful route to realize designer moiré systems with programmable quantum properties, opening new opportunities for optoelectronic applications.

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The electronic properties of bilayer moiré systems depend crucially on the periodicity, symmetry, topology, and dimensionality of the underlying microscopic structure [1]. In

particular, strain effects are considerably more pronounced in bilayers compared to individual layers, and can substantially alter the original moiré superlattice [2,3]. Meanwhile, to minimize the stacking energy, the superlattice tends to spontaneously rearrange into a domain-wall pattern, particularly at small twist angles [4,5]. This atomic reconstruction phenomenon, along with random strain generated during fabrication, have been identified as major factors contributing to complex spatial variations in realistic moiré samples [6–10], which in turn dramatically affect their transport and optical behaviors [11,12]. Enhanced control over these two structural factors offers great potential for engineering customized moiré superlattices.

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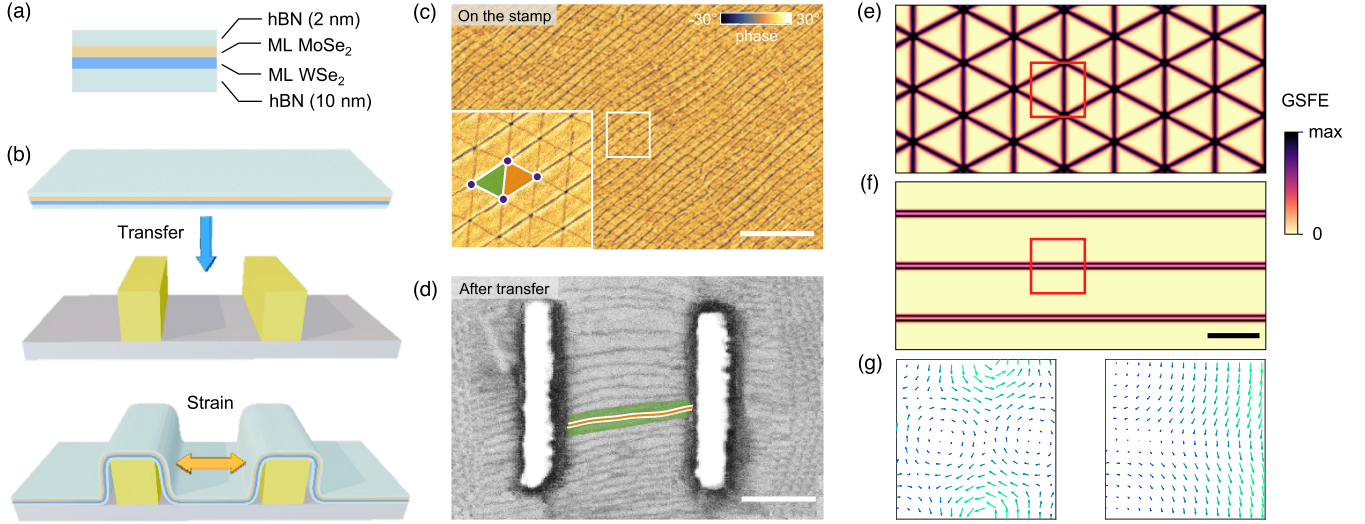


FIG. 1. (a) Schematic cross-section of a representative MoSe₂-WSe₂ heterobilayer sample encapsulated in thin hBN layers, with the thickness of the top and bottom hBN given in parentheses. (b) Illustration of the experimental method to locally induce uniaxial strain on the heterobilayer. After assembly using a polymer stamp, the flat sample is transferred onto a patterned substrate featuring parallel elongated mesas. Due to strong adhesion, the sample conforms to the mesa contours, resulting in uniaxial strain between the mesas. (c) TFM phase image of the heterobilayer before transfer, showing a homogeneous region that features a 0D periodic lattice. The inset highlights the alternating triangular pattern of the domains, which correspond to the two energetically favorable stackings R_h^M and R_h^X (green and orange, respectively). (d) SEM image of the same region as in (c) after transfer, revealing the formation of a 1D array consisting of elongated R_h^M domains (green) separated by narrow domain walls of R_h^X stacking (orange). (e), (f) Calculated generalized stacking fault energy maps for a triangular moiré pattern with $\theta = 0.4^\circ$ and the corresponding 1D pattern. (g) Left (right) panel: deformation field in the area indicated by the red square in (e), (f). Scale bars: 500 nm in (c), (d) and 50 nm in (f).

In this Letter, we demonstrate control of the moiré patterns and exciton properties in MoSe₂-WSe₂ heterobilayers by exploiting the interplay of uniaxial strain and atomic reconstruction. Figure 1(a) illustrates schematically the cross-sectional structure of a representative sample fabricated for strain control. By a layer-by-layer all-dry stamping technique, MoSe₂ and WSe₂ monolayers synthesized by chemical vapor deposition were aligned to around 0° (R-type) and encapsulated by hexagonal boron nitride (hBN) layers. The 2-nm-thick top hBN layer enables direct visualization of moiré superlattice via nanoimaging, while the 10-nm-thick bottom hBN layer ensures both optical quality [13] and mechanical coupling to the patterned substrate. As illustrated in Fig. 1(b), to apply local strain to the heterobilayer, we designed a substrate patterned with parallel pairs of elongated mesas and transferred the sample onto it (Supplemental Material [14] Note 1). Due to strong van der Waals forces, the originally flat sample conforms to the contours of the patterned substrate after the transfer, causing the region between the mesa pair to stretch [35]. Therefore, a local uniaxial tensile strain of $\sim 0.1\%$ is applied to the heterobilayer (Supplemental Material [14] Fig. S1).

To reveal the effect of strain on the moiré structure, we performed real-space nanoimaging on the heterobilayer before and after transfer. Initially, when the heterobilayer was on the stamp prior to capping with the bottom hBN

layer, we mapped its structure using torsional force microscopy (TFM). This atomic force microscopy-based technique offers a non-destructive method to visualize moiré superlattices with lateral resolution below 1 nm [21]. Figure 1(c) presents a typical TFM phase image recorded in the core of the heterobilayer. We observe a uniform lattice formed by triangular zero-dimensional (0D) domains approximately 60 nm in size, consistent with theoretical predictions of lattice reconstruction in R-type heterobilayers [22,23]. Note that this 0D lattice actually consists of two alternating energetically favored stackings R_h^M and R_h^X (highlighted in green and orange) [5,36]; however, TFM can only distinguish the boundaries between them, appearing as thin lines of darker contrast. After characterization of the entire heterobilayer with TFM, we transferred the sample by aligning the homogeneous 0D regions with the mesa pairs to avoid random strain, which can also change the reconstruction pattern [12].

Following the transfer process, we carried out electron channeling contrast imaging in scanning electron microscopy (SEM) to visualize the heterobilayer. In contrast to TFM, this technique differentiates the two optimal stackings R_h^M and R_h^X according to their distinct electron channeling conditions [37]. Figure 1(d) shows a SEM image of the same region as Fig. 1(c) for comparison. We find that the applied uniaxial strain causes the triangular domains of R_h^M stacking to expand and merge into an array

of elongated domains (example marked in green), bridging the two mesas and maintaining a width of ~ 60 nm. Meanwhile, the R_h^X stacking collapsed into few-nanometer-wide double domain walls that separate the R_h^M domains on both sides (marked in orange). This local transformation from 0D lattice into 1D array also occurred in other mesa pairs of the same sample as well as other samples (Supplemental Material [14] Fig. S5).

To explain the mechanism of strain-induced 1D wire array formation, we adopt the theoretical model described in Refs. [12,23], explicitly including the effect of uniaxial strain. It is established that the formation of a 1D array consisting of elongated domain walls can become energetically more favorable as compared to a triangular network with nodal domain walls [38–40]. In our case, we demonstrate that uniaxial strain decreases the symmetry and effectively modifies the boundary conditions in simulations, leading to a periodic 1D superlattice in the energetically optimal configuration, as indicated by generalized stacking fault energy (GSFE) or adhesion energy maps [22,23] in Figs. 1(e) and 1(f). The simulations show the formation of elongated domains of R_h^M stacking and double domain walls with the core exhibiting R_h^X stacking, consistent with the experimental results. Figure 1(g) illustrates the corresponding displacement fields for these two configurations. Further details of the theoretical model and simulations are provided in Supplemental Material [14] Note 2.

We now explore the optical properties of the 1D lattices. *Ab initio* calculations show that the interlayer exciton energies for R_h^X and R_h^M stackings are 1.33 and 1.45 eV, respectively [41]. As illustrated in Fig. 2(a), this large

energy offset of ~ 120 meV strongly modulates the potential landscape of the 1D lattice, enabling efficient trapping of interlayer excitons within the domain walls of R_h^X . Compared with free interlayer excitons in R_h^X stacking, the 1D-confined excitons not only retain a permanent out-of-plane electric dipole moment, as electrons and holes reside in different layers, but are also expected to change their optical selection rule from circular to linear polarization along the domain wall. This arises because the 1D confinement breaks the C_3 symmetry of the exciton wave function and hybridizes the two valleys via electron-hole exchange interaction [12].

To verify this, we perform linear polarization-resolved PL measurements at 4 K. Figures 2(b) and 2(c), respectively, show an optical micrograph of the sample after applying strain and the map of the degree of linear polarization (P_l) in the region of mesa pairs. Here, P_l is defined as $P_l = (I_H - I_V)/(I_H + I_V)$, where I_H (I_V) represents the integrated PL intensity detected under the polarization along the horizontal (vertical) direction of the optical micrograph in Fig. 2(b). We observe that all strain-induced 1D arrays exhibit near-perfect linear polarization ($|P_l| > 0.9$) with the angle aligned along the domain wall axis and perpendicular to the corresponding mesa orientation. This determined linear polarization strongly contrasts with the results from randomly strained heterobilayers, where the occurrence of linearly polarized interlayer exciton PL is arbitrary and the $|P_l|$ values are relatively low [42,43].

Figure 2(d) presents linear polarization-resolved PL spectra of 1D-confined interlayer excitons under $1 \mu\text{W}$ excitation, with an emission peak of 7 meV linewidth

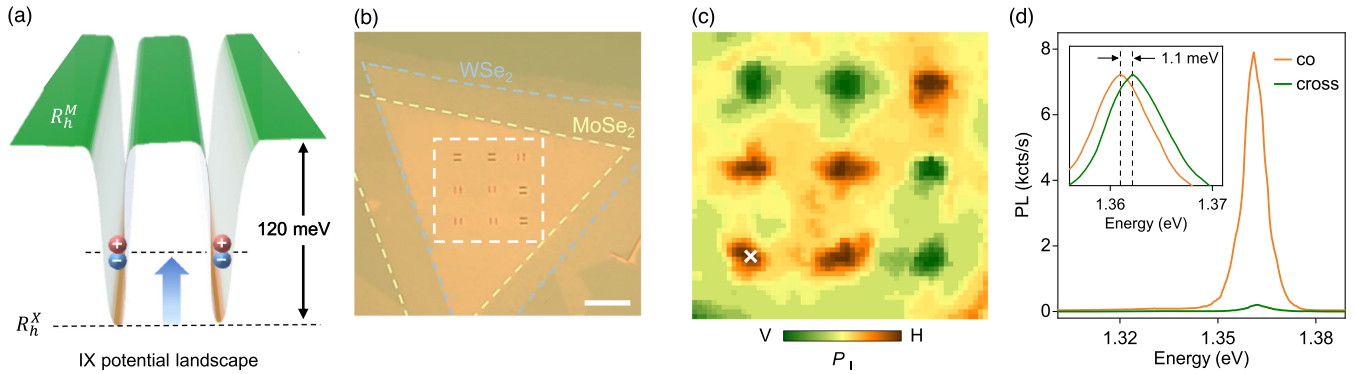


FIG. 2. (a) Schematic of the interlayer exciton (IX) potential landscape in a 1D array region. The narrow width of R_h^X domain walls and the high energy barrier from adjacent R_h^M domains confine interlayer excitons within the domain walls, resulting in a blue-shift of their energy compared to the free exciton energy of R_h^X stacking (arrow). (b) Optical micrograph of the sample after transfer onto a patterned substrate with nine pairs of horizontally or vertically oriented mesa pairs. The dashed yellow and blue lines indicate the locations of monolayer MoSe₂ and WSe₂, respectively. The scale bar is 5 μm . (c) 2D spatial map of linear polarization (P_l) of interlayer exciton emission within the area marked by the white box in (b). Orange and green, respectively, indicate horizontal (H) and vertical (V) polarization, defined by the orientation of the micrograph in (b). The hot spots align with the positions of mesa pairs, with polarization matching the domain wall axis orthogonal to the elongated mesas. (d) Linear polarization-resolved PL spectra of 1D-confined interlayer excitons, contrasting co- and cross-polarized components relative to the domain wall axis, acquired at the location marked by the cross symbol in (c). The inset shows normalized spectra highlighting the energy splitting of linearly polarized interlayer exciton states.

centered at ~ 1.36 eV. The peak energy exhibits a pronounced 30 meV blue shift compared to free interlayer excitons in 2D R_h^X domains (Supplemental Material [14] Fig. S6), consistent with the size-quantization effect in the potential landscape described in Fig. 2(a). Notably, the co- and cross-polarized components display a nonnegligible splitting of 1.1 meV [inset of Fig. 2(d)], contrasting with the valley-degenerate circular polarization of free interlayer excitons. With increased excitation power, a secondary peak emerges 20 meV above the primary emission, with identical polarization characteristics and energy splitting (Supplemental Material [14] Figs. S10 and S11). This power-activated higher-energy feature is reminiscent of excited excitonic states observed in the systems with strong confinement [44].

To gain insight into these behaviors of 1D-confined interlayer excitons, we develop a microscopic theory that takes into account the long-range electron-hole exchange interaction that underlies the exciton fine structures [29,30], and model the 1D confinement of the exciton center-of-mass as an infinite rectangular quantum well (see Supplemental Material [14] Note 3 for details). Within this framework, the splitting of the 1D-confined exciton doublet is given by $|\Delta_{xy}| = \pi \hbar \Gamma_0 / (\epsilon_{\text{eff}} q w)$, where Γ_0 is the

exciton radiative decay rate of “bulk” 2D interlayer exciton and $q = \omega/c$ is the photon wave vector at the exciton resonance frequency ω . Using the parameters of our structure, we find that $|\Delta_{xy}| \sim 1$ meV is readily achievable. Moreover, the ordering of the polarized eigenstates is consistent with experiment: the lower-energy sublevel is polarized parallel to the 1D domain walls. At the experimental temperature of 4 K, the upper sublevel is negligibly thermally populated due to the sizable splitting, accounting for the observed near-unity linearly polarized PL from the ensemble of domain walls. In addition, electrodynamic effects associated with the proximity of gold to the domain walls may further enhance the degree of linear polarization [45].

Harnessing the subnanometer resolution of TFM imaging, we are able to precisely measure the R_h^X -stacked domain wall width w . Figures 3(a)–(c) display representative TFM phase images of 1D arrays from different mesa pairs, with w of 15, 7, and 3 nm, respectively. From the average width of 27 domain-wall arrays in three heterostructure samples (nine mesa pairs per sample), we infer an inverse relationship between the pretransfer twist angle θ and poststrain domain wall width: smaller θ values yield broader domain walls. This trend is further validated by

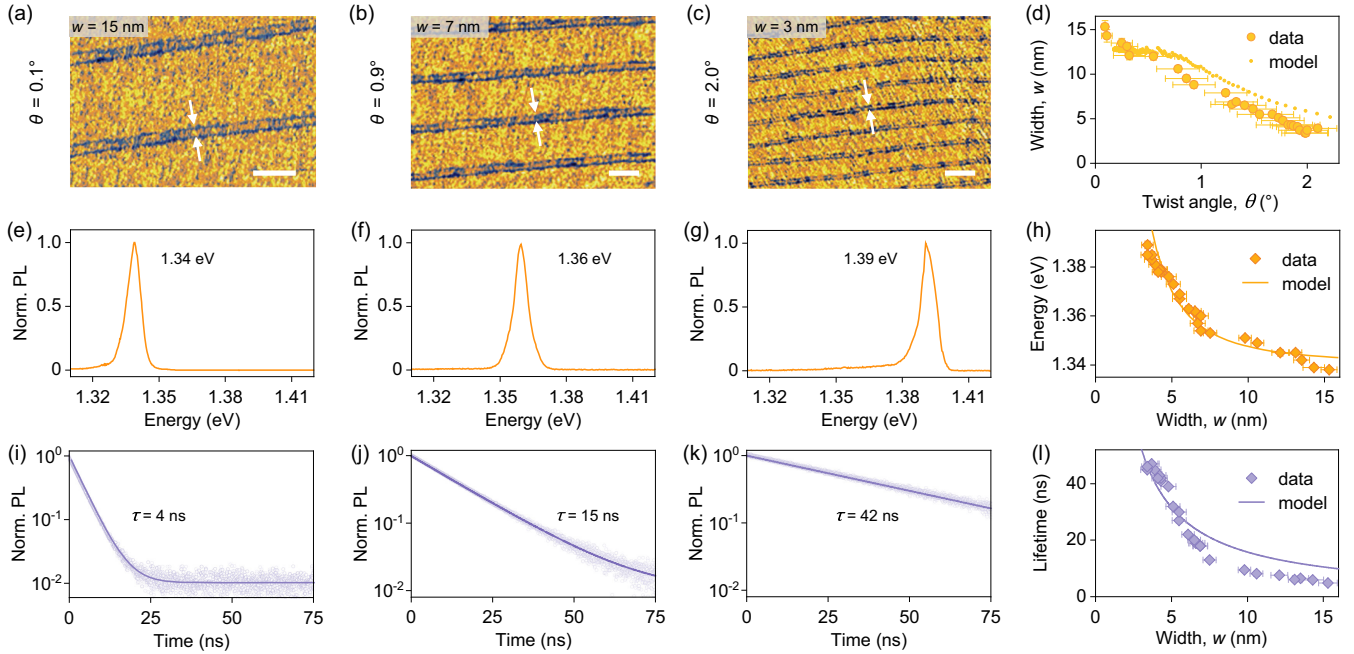


FIG. 3. (a)–(c) TFM phase images of three 1D arrays with domain wall width w of 15 nm (a), 7 nm (b), and 3 nm (c), where w is defined as the width of the high-contrast region in the images (indicated by the arrows). The corresponding local twist angles θ before strain are shown on the left (calculated from the average triangular domain size). The scale bars are 50 nm, 20 nm and 15 nm, respectively. (d) Large dots: plot of w as a function of θ , summarizing all strain-induced 1D arrays in three samples. Small dots: simulated domain wall width based on the atomic reconstruction model (see Supplemental Material [14] Note 2). (e)–(l) Correlation of 1D-confined interlayer excitons with domain wall width w . Normalized PL spectra (e)–(g) and PL decay (i)–(k) acquired from the 1D arrays shown in (a)–(c). The energy position of peak maximum obtained from Lorentzian fitting is indicated in each PL spectrum. The PL decay traces are fitted with single exponential functions (lines) with lifetimes indicated on the side. (h), (l) Dependence of interlayer exciton energy (h) and lifetime (l) on w . Solid lines represent calculated results obtained from the model described in Supplemental Material [14] Notes 3 and 4.

statistical mapping of w versus θ for all strain-controlled 1D lattices [Fig. 3(d)]. We qualitatively reproduce this dependence with our theoretical model (small dots; see Supplemental Material [14] Note 2 for details). Note that the numerical simulations overestimate the domain wall widths at large angles. We attribute this discrepancy to an underestimated GSFE for intermediate stacking configurations in the simulations, resulting in enlarged domain wall areas [7].

In the following, we correlate the double domain wall width and the optical response of 1D-confined interlayer excitons. Figures 3(e)–3(g) and 3(i)–3(k) present PL spectra and time-resolved PL dynamics acquired from the regions in Figs. 3(a)–3(c). The widest domain wall ($w = 15$ nm) exhibits a PL peak at 1.34 eV [Fig. 3(e)], blue shifted by 10 meV relative to the free interlayer excitons. The corresponding monoexponential PL decay yields a 4 ns lifetime [Fig. 3(i)], slightly longer than the 2.5 ns characteristic of unconfined excitons (Supplemental Material [14] Fig. S6). As summarized in Figs. 3(h) and 3(l), progressive narrowing of w induces gradual spectral blue-shift and lifetime extension, reaching maxima of 1.39 eV (a 60 meV shift) and 42 ns for $w = 3$ nm [as exemplified in Figs. 3(g) and 3(k)]. Meanwhile, the co-/cross-polarization splitting Δ_{xy} decreases from 1.7 to 0.3 meV (Supplemental Material [14] Fig. S9). These trends in Δ_{xy} , lifetime τ , and emission energy E are well captured by our theoretical model of confined excitons, with scaling relations $\Delta_{xy} \propto 1/w$, $\tau \propto 1/w$, and $E - E_0 \propto 1/w^2$, where E_0 denotes the free interlayer exciton energy (Supplemental Material [14] Notes 3 and 4).

Owing to the permanent out-of-plane electric dipole, external electric field can efficiently shift the energy of interlayer excitons through the quantum-confined Stark effect [46]. To examine this tunability for the 1D-confined excitons, a dual-gated sample containing strain-induced 1D arrays was fabricated. Figure 4(a) displays the linear energy shift of the 1D-confined interlayer excitons with electric field, yielding a dipole moment of 0.65 e nm according to the linear slope, in good agreement with *ab initio* estimation of layer separation for R_h^X stacking [47]. Remarkably, over an energy range of ~ 100 meV (from 1.31 to 1.41 eV), the 1D-confined excitons preserve near-perfect linear polarization [Fig. 4(b)], achieving unprecedented *in situ* tunability while maintaining anisotropic emission—a combination unmatched in other 1D quantum systems [48]. In addition to the energy shifts, the PL intensity also changes with the electric field, since the electron and hole wave functions and thus the optical transition matrix elements of the 1D-confined interlayer excitons change, resulting in an increase in Γ_0 [49,50]. Consistently, the energy splitting between the two linearly polarized states increases from 0.8 to 1.6 meV over the electric field range [Fig. 4(c)], highlighting again the high tunability of the system. It has been recently reported that under high fields (above 1 V/nm) the

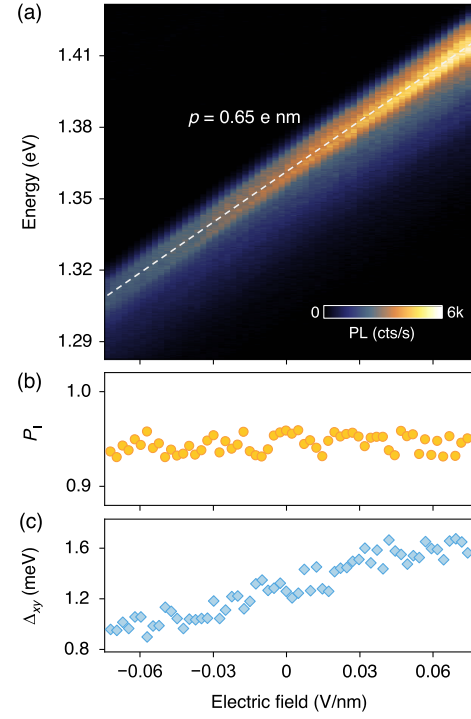


FIG. 4. (a–c) Evolution of the PL spectra (a), the degree of linear polarization P_1 (b) and the energy splitting Δ_{xy} between linearly polarized states (c) as a function of out-of-plane electric field. Data recorded from a 1D array in a dual-gated sample, which includes additional top, bottom, and contact electrodes made from few-layer graphene.

domain walls in twisted homobilayers are possible to move due to ferroelectric effects [51]. The related field-driven domain wall dynamics in heterobilayers as well as their implications for interlayer excitons cannot be excluded and remain to be addressed in future work.

In conclusion, we realize strain-induced atomic reconstruction in MoSe_2 - WSe_2 moiré heterobilayers that stabilizes parallel domain-wall arrays and thereby forms quasi-1D confinement channels for interlayer excitons. Correlative nanoscale imaging and polarization-resolved cryogenic PL consistently indicate confinement-induced symmetry lowering, accompanied by a fine-structure splitting consistent with valley mixing in the 1D geometry. By varying the interlayer twist angle, we tune the channel width and the resulting exciton energy and dynamics. In dual-gated devices, we further achieve large Stark tunability of the trapped interlayer excitons. These results establish a deterministic route to realize 1D dipolar excitons in moiré systems, complementary to recent gate- and ferroelectric-domain-wall-defined 1D exciton platforms [52,53], and enable controlled access to strongly interacting regimes of 1D dipolar excitons, for example, the density-driven crossover from a Tonks–Girardeau gas to a charge-density-wave state [54].

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Data availability—All data for this Letter are available at Zenodo [55].

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